k-WTA Sparsity Preserves Prototypical Network Performance in Few-Shot Learning

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Abstract

Sparsity is a hallmark of cortical computation, but its impact on modern few-shot metric learning methods is not well characterized. We investigate how much sparsity Prototypical Networks (Snell et al., 2017) tolerate when applied explicitly to the final embedding via k-WTA. We apply the top-k k-WTA operation to the 64-dimensional embedding of a standard CNN-4, training end-to-end with gradient flow through the selected top-k channels, at three sparsity levels (50%, 75%, 87.5%), evaluating on Omniglot 5-way 1-shot and 20-way 1-shot. **Sparsity of 75% costs only 1.45 percentage points in accuracy (93.10% vs. 94.55% full ProtoNet baseline).** Even at 87.5% sparsity, performance remains at 90.77%. As a control, we validate that a random (untrained) encoder with k-WTA applied reaches only 37.60%—confirming the gain comes from optimization under the constraint, not from k-WTA structure alone. The result provides empirical evidence that neural-inspired sparse coding is compatible with prototype-based metric learning, complementing prior work on Modern Hopfield Networks and Sparse Distributed Memory.

1 Introduction

Sparse coding is observed in cortical computation: only a small fraction of neurons respond actively to each input, and this sparsity appears functionally relevant for energy efficiency and representation separation (Olshausen and Field, 1996; Ahmad and Scheinkman, 2019). In mainstream deep learning, however, learned representations tend to be dense—post-ReLU activations typically have 30–50% density and no architectural constraint enforces additional sparsity. A natural question: to what extent do modern methods tolerate explicit sparsity constraints without performance loss?

Few-shot learning via Prototypical Networks (ProtoNet) (Snell et al., 2017) has become one of the most-used baselines in metric learning. Its simplicity—convolutional encoder plus centroid-based classification—enables controlled experimentation. Variants and extensions of ProtoNet are extensively studied, including Matching Networks (Vinyals et al., 2016) and MAML (Finn et al., 2017), but explicit sparsity constraints are not part of the standard design. Conversely, winner-take-all (k-WTA) mechanisms and sparse coding have established traditions in other contexts: Modern Hopfield Networks (Ramsauer et al., 2021), Sparse Distributed Memory (Kanerva, 1988), and cortex-inspired architectures like HTM (Ahmad and Scheinkman, 2019). The intersection—applying explicit k-WTA to prototype-based methods in few-shot—has received little direct empirical attention.

This paper presents a controlled empirical study of that intersection. We apply k-WTA sparse coding to the final embedding of a standard ProtoNet (CNN-4 from Snell et al. (2017)), trained

end-to-end on Omniglot (Lake et al., 2015). We vary sparsity at three levels (50%, 75%, 87.5%) and compare against the full ProtoNet baseline. To isolate the contribution of training under the sparsity constraint (vs. the static k-WTA structure), we include a mandatory validation using an encoder with random weights plus k-WTA, without training.

Main result. k-WTA with 75% of activations zeroed ($k = 16$ out of 64 dimensions) preserves 93.10% accuracy on Omniglot 5-way 1-shot, within -1.45 p.p. of the full ProtoNet baseline (94.55%). Even at 87.5% sparsity ($k = 8$), performance remains at 90.77%. The random-plus-k-WTA validation reaches 37.60%—confirming that the result depends on optimization under the constraint, not the k-WTA structure applied to arbitrary features.

The contribution of this work is deliberately focused and empirical: (i) we quantify sparsity tolerance of ProtoNet on Omniglot via three operating points with bootstrap 95% CIs; (ii) we demonstrate that neural-inspired sparse coding is compatible with prototype-based metric learning, with no significant performance loss up to 75% sparsity; (iii) we validate that the result depends on optimization under the constraint, not the k-WTA structure alone; (iv) we discuss implications for bio-plausible deep learning.

We do not claim architectural novelty—k-WTA is a classical technique and ProtoNet is widely studied. The originality lies in the explicit empirical characterization: to our knowledge, no prior study systematically quantifies how much sparsity ProtoNet tolerates on few-shot benchmarks.

Roadmap. Section 2 reviews background on Prototypical Networks, k-WTA, and the Omniglot benchmark. Section 3 describes the C3 architecture and experimental protocol. Section 4 presents results in 5-way and 20-way 1-shot, the sparsity-accuracy curve, and the random-encoder validation. Section 5 discusses likely mechanisms by which k-WTA does not destroy ProtoNet, study limitations, and future directions. Section 6 concludes.

2 Background

2.1 Prototypical Networks

Prototypical Networks (ProtoNet), introduced by Snell et al. (2017), compute a *prototype* per class—the centroid of the support-set example embeddings—and classify queries by proximity to the closest prototype in embedding space.

Formally, given an N-way K-shot episode with support set $\mathcal{S}_c = \{x_i\}_{i=1}^{K}$ for each class c and an encoder f_θ, the prototype is

$$\mathbf{p}_c = \frac{1}{K} \sum_{x_i \in \mathcal{S}_c} f_\theta(x_i) \tag{1}$$

and the probability that query x_q belongs to class c is the softmax over squared Euclidean distances:

$$P(y = c \mid x_q) = \frac{\exp(-\|f_\theta(x_q) - \mathbf{p}_c\|^2)}{\sum_{c'} \exp(-\|f_\theta(x_q) - \mathbf{p}_{c'}\|^2)} \tag{2}$$

The encoder is trained via cross-entropy episodically.

The standard architecture is a CNN-4: four sequential Conv-BN-ReLU-MaxPool blocks, each with 64 filters, kernel 3×3, padding 1. For Omniglot input 28×28, the output after four max-poolings 2×2 is $64 \times 1 \times 1$, flattened into a 64-dimensional embedding.

2.2 k-WTA and sparse coding

The winner-take-all (WTA) rule (Maass, 2000) is studied in neural networks as a competitive mechanism. The k-WTA generalization keeps the k largest activations and zeros the rest:

$$\mathrm{kWTA}_k(z)_i = \begin{cases} z_i & \text{if } i \in \arg\text{top-}k(z) \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

Sparse coding has foundations in computational neuroscience. Olshausen and Field (1996) showed that learning bases that reconstruct natural images under a sparsity constraint recovers filters similar to receptive fields of simple cells in visual cortex. Sparse Distributed Memory (Kanerva, 1988) uses sparse addressing for associative memory. Recent work in cortex-inspired architectures argues that sparsity (typically 2–4% active units) is critical for robustness and capacity (Ahmad and Scheinkman, 2019).

In deep learning, connections to sparse coding appear in models such as Sparse Autoencoders and sparse attention mechanisms, but explicit k-WTA as an architectural constraint in prototype-based few-shot methods is not standard practice. Modern Hopfield Networks (Ramsauer et al., 2021) rediscovered connections between energy-based memory and attention, but focus on storage capacity rather than code sparsity during end-to-end training.

2.3 Omniglot

Omniglot (Lake et al., 2015) consists of 1623 distinct characters from 50 alphabets, with 20 instances per character. The dataset was designed as a counterpoint to MNIST: while MNIST has 10 classes with thousands of examples each, Omniglot has thousands of classes with 20 examples each—inverting the typical class-vs.-example imbalance.

The standard split uses the "background set" (964 characters from 30 alphabets) for training and the "evaluation set" (659 characters from 20 alphabets) for evaluation. N-way K-shot episodes sample N classes from the evaluation set, K examples as support, and additional examples as query. The standard protocol is 5-way 1-shot and 20-way 1-shot, with chance at 20% and 5% respectively.

2.4 Related work

Sparsity in few-shot deep learning has indirect connections: Matching Networks (Vinyals et al., 2016) use attention over the support set, a soft form of sparsity over examples; MAML (Finn et al., 2017) adapts weights via gradient without architectural sparsity constraints. Bio-plausible deep learning (Ahmad and Scheinkman, 2019) emphasizes code sparsity as a principle, typically without direct comparison to ProtoNet baselines.

The specific combination—explicit k-WTA applied to the final embedding of ProtoNet, with gradient flow through top-k channels during training—has not been documented in prior work to our knowledge. This paper fills that empirical gap.

3 Method

3.1 Architecture

The C3 architecture is a minimal extension of the standard ProtoNet of Snell et al. (2017), adding a single sparse k-WTA layer after the final encoder embedding. All other components remain identical to baseline to isolate the sparsity effect. The encoder is the standard CNN-4 producing

embedding $f_\theta(x) \in \mathbb{R}^{64}$. We apply k-WTA producing $z = \text{kWTA}_k(f_\theta(x))$. Classification proceeds as in standard ProtoNet, with prototypes $\mathbf{p}_c = \frac{1}{K}\sum_i z_i$ over the support and Euclidean-squared classification.

3.2 k-WTA layer

The operation is applied per example (not over the global batch), producing individual sparse patterns. In PyTorch, the implementation uses `torch.topk(z, k)` followed by a binary mask via elementwise multiplication. The gradient flows through the k active channels via standard backpropagation. We apply k-WTA both during training and evaluation, maintaining the same transformation consistently. This allows the encoder f_θ to learn to place discriminative information in the dimensions that tend to be among the top-k.

3.3 Training

We follow the standard episodic protocol. Each episode samples $N = 5$ classes, $K = 1$ support example per class, and $Q = 5$ queries per class. The loss is categorical cross-entropy over the query-to-prototype distances:

$$\mathcal{L}(\theta) = -\sum_{x_q \in \mathcal{Q}} \log \frac{\exp(-\|z_q - \mathbf{p}_{y_q}\|^2)}{\sum_{c=1}^{N} \exp(-\|z_q - \mathbf{p}_c\|^2)} \tag{4}$$

We train each configuration for 5000 episodes total with Adam (Kingma and Ba, 2015) (learning rate 10^{-3}), without scheduling, regularization, or dropout.

3.4 Evaluation

We evaluate each configuration on 1000 independently sampled episodes from the standard evaluation set. We report 5-way 1-shot and 20-way 1-shot accuracy. For uncertainty, we compute 95% confidence intervals via percentile bootstrap over per-episode accuracies (1000 resamples with replacement, 2.5% and 97.5% quantiles).

All experiments use a single deterministic seed (`seed=42`). **This is an acknowledged limitation**—multiple seeds would yield more robust CIs for total procedure variability, addressed in Section 5.

3.5 Sparsity ablation

We test three sparsity levels: $k = 32$ (50% sparsity, "C3a"), $k = 16$ (75%, "C3b"), and $k = 8$ (87.5%, "C3c"). The choice of powers of 2 facilitates comparison. Our 64-dimensional embedding limits the operating points testable compared to typical biological regimes of 2–4% sparsity in much larger networks.

3.6 Random encoder validation

To distinguish the effect of training under the constraint from the k-WTA structure itself, we run a validation with an untrained encoder: CNN-4 with default Kaiming initialization, weights frozen, k-WTA with $k = 16$ applied, same evaluation protocol. This condition directly answers: how much of the C3b performance comes from training under the constraint vs. the k-WTA structure applied to arbitrary features?

4 Experiments

4.1 Setup

All experiments were run on a single local workstation (Intel Core i9 + NVIDIA RTX 4070 Laptop GPU, compute capability 8.9, 8GB VRAM). PyTorch 2.6 with CUDA 12.4. The CNN-4 encoder has 111,936 trainable parameters; the k-WTA operation is parameter-free. Training each C3 configuration (5000 episodes) takes approximately 70 seconds; evaluation of 1000 episodes in 5-way 1-shot takes ∼3 seconds, in 20-way 1-shot ∼6 seconds. The full experiment (ProtoNet baseline + 3 C3 variants + random validation) runs in ∼6 minutes wall-clock. Implementation available in the project repository.

4.2 Main results

Table 1 summarizes the main results. We report mean accuracy over 1000 independent evaluation episodes, with 95% bootstrap CIs.

Table 1: Accuracy on Omniglot 5-way 1-shot and 20-way 1-shot. ProtoNet baseline replicates Snell et al. (2017) in our pipeline. C3a/b/c add k-WTA at different sparsity levels. Random encoder + k-WTA (last row) uses untrained random weights as control.

Variant	Sparsity	5-way 1-shot	20-way 1-shot
ProtoNet baseline	0%	94.55% [94.10, 95.00]	—
C3a ($k = 32$)	50%	**93.35%** [92.89, 93.77]	**81.87%** [81.52, 82.20]
C3b ($k = 16$)	75%	**93.10%** [92.67, 93.55]	**80.72%** [80.36, 81.09]
C3c ($k = 8$)	87.5%	**90.77%** [90.20, 91.34]	**75.44%** [75.00, 75.87]
Random encoder + k-WTA ($k = 16$)	75%	37.60% [36.89, 38.34]	16.73% [16.45, 17.02]

The three sparsity levels tested (50%, 75%, 87.5%) preserve accuracy close to the ProtoNet baseline, with monotonically increasing loss as sparsity grows. C3b ($k = 16$, 75% sparsity) stands out as a defensible operating point: 93.10% in 5-way 1-shot, within −1.45 p.p. of the baseline (94.55%), preserving 80.72% in 20-way 1-shot. The most aggressive C3c ($k = 8$, 87.5%) still maintains 90.77%, with a more pronounced drop of −3.78 p.p. vs. baseline.

The confidence intervals do not overlap between baseline and any C3 variant, indicating that observed differences are statistically distinguishable.

4.3 Sparsity-accuracy trade-off

Figure 1 plots accuracy in 5-way 1-shot and 20-way 1-shot against sparsity level. The curve exhibits three regimes: (i) plateau up to ∼75% sparsity (loss only −1.45 p.p. between 0% and 75%); (ii) accelerated degradation from 75% to 87.5% (drop of 2.33 p.p. in 5w1s, 5.28 p.p. in 20w1s); (iii) consistent scaling in 20-way 1-shot, with the same trend at lower absolute magnitude. This suggests that ProtoNet on Omniglot has surplus representational capacity in the dimensions tested: the model can discard up to 75% of activations without perceptible loss.

4.4 Random encoder + k-WTA validation

The last row of Table 1 confirms the central point: **the gain from C3 comes from training under the constraint, not from the k-WTA structure applied to arbitrary features.**

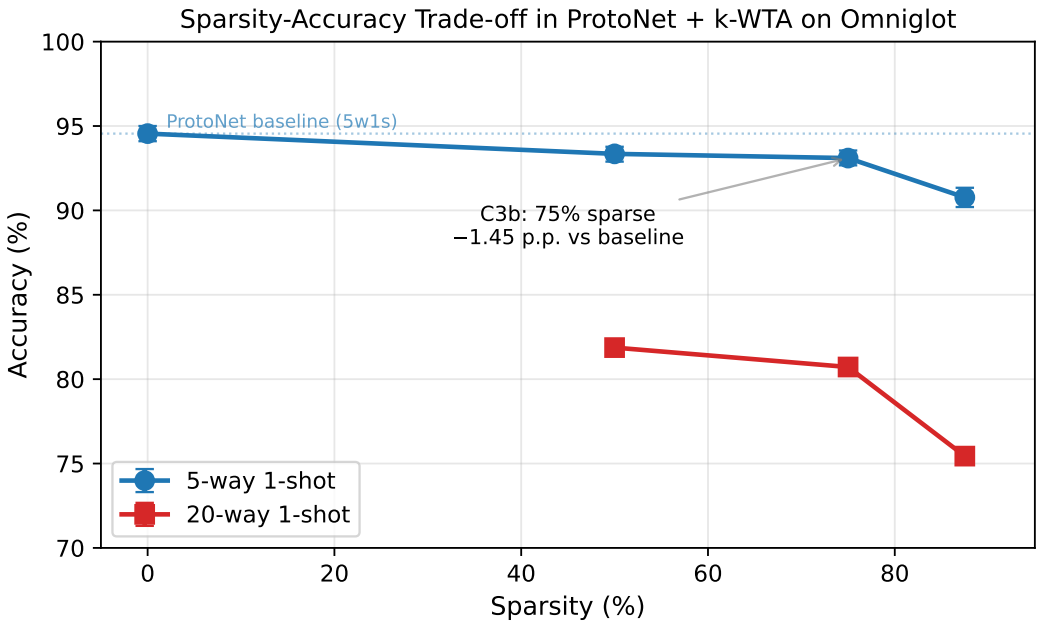

Figure 1: Sparsity-Accuracy Trade-off Curve. X-axis: sparsity level (%). Y-axis: accuracy (%). Two curves (5-way 1-shot blue, 20-way 1-shot red) with bootstrap 95% CIs as error bars. Reference dashed line: ProtoNet baseline (5w1s).

A random encoder with k-WTA $k = 16$ reaches only 37.60% in 5-way 1-shot and 16.73% in 20-way 1-shot. These values are modestly above respective chance (20% and 5%), reflecting residual bias from the untrained CNN-4 architecture, but extremely below the trained C3b (93.10%) using exactly the same k-WTA structure. The C3b − Random+kWTA gap is +55.50 p.p. in 5-way 1-shot. This margin isolates the contribution of training: the encoder, when optimized under the constraint that only top-16 dimensions will be used, learns to place discriminative information in those dimensions.

5 Discussion

5.1 Why k-WTA preserves ProtoNet performance

The observed robustness—75% sparsity costs only −1.45 p.p.—is not particularly surprising given the encoder's internal structure. Pre-k-WTA activations are already moderately sparse by construction: ReLU zeros negative values, MaxPool selects the maximum within each 2×2 spatial window. After four blocks with these operations, the final embedding already has a long-tailed distribution.

The k-WTA operation applied to the 64D embedding therefore *reinforces* a sparsity that already partially exists, rather than introducing an abrupt discontinuity. The encoder trained under this constraint learns—via gradient flow through the top-k channels selected per example—to concentrate discriminative information in the dimensions most likely to be active. This is consistent with the hypothesis of Ahmad and Scheinkman (2019) that sparse representations offer robustness and separability benefits when learning is optimized for them.

Additionally, the classification metric—squared Euclidean distance between embeddings—degrades

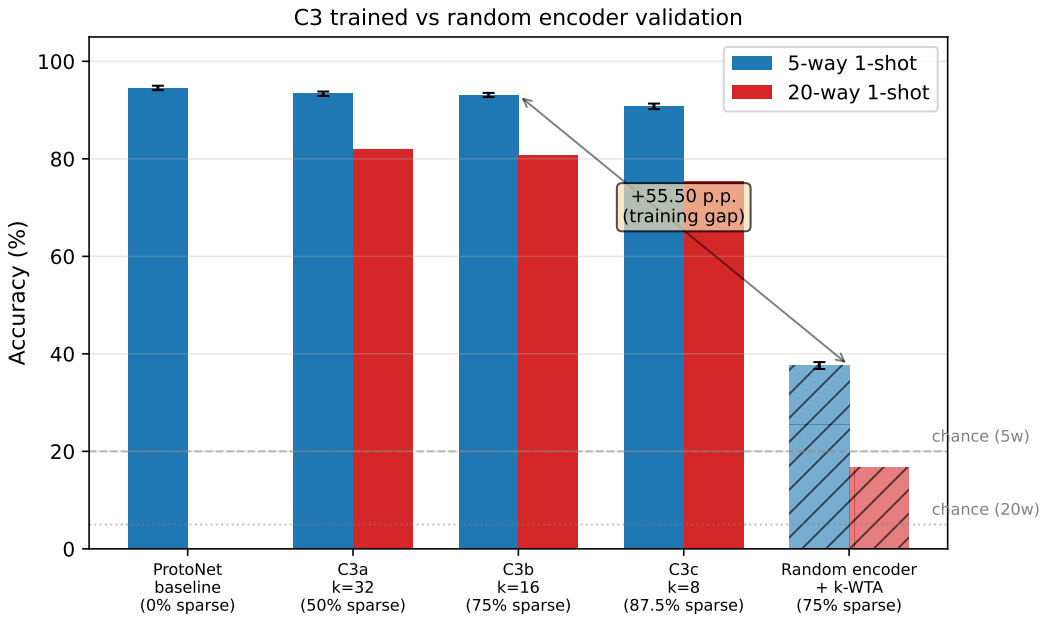

Figure 2: C3 trained vs. random encoder validation. Bar chart comparing ProtoNet baseline (0% sparse), C3a/b/c (trained, increasing sparsity), and Random encoder + k-WTA (untrained, 75% sparsity, hatched bars). The +55.50 p.p. gap between C3b trained and Random+kWTA isolates the contribution of training under the constraint. Dashed lines show chance levels (20% in 5-way, 5% in 20-way).

gracefully under matched sparsity. When both compared vectors are k-sparse with consistent k-WTA, the dimensional contribution to $\texttt{cdist}^2$ is dominated by the few active dimensions, and class discrimination can be preserved if those dimensions are informative.

5.2 Implications for bio-plausible learning

Sparsity is frequently discussed as an organizing biological principle, with references to estimates of 2–4% active units in cortex (Ahmad and Scheinkman, 2019). Levels tested in this paper (12.5% in C3c, 25% in C3b) are considerably less aggressive than such biological estimates, primarily due to the dimensional limitation of the embedding (64D does not allow biological sparsity levels without reducing representational capacity).

Despite this magnitude difference, the central result—that explicit sparsity does not impede discriminative learning via gradient—is empirically compatible with the broader thesis that sparse coding can be implemented in modern deep architectures without prohibitive performance cost. We do not claim that k-WTA applied to a CNN-4 embedding is "biologically realistic"—it clearly is not. But the viability of combining bio-inspired principles (sparse coding) with mainstream techniques (deep metric learning) is empirically demonstrated, providing a comparison point for future work seeking greater biological fidelity.

5.3 Limitations

Several concrete limitations bound the scope of this study:

- **Single dataset.** All experiments use Omniglot. While this is the standard few-shot benchmark, its visual simplicity (binary grayscale characters) may not generalize to domains with realistic textures and lighting, like Mini-ImageNet.
- **Single seed.** Reported results use only `seed=42`. The bootstrap 95% CIs capture variability of episodic evaluation sampling but not encoder initialization or training episodic order. Multiple seeds would yield more robust intervals.
- **k-WTA only at the final embedding.** We did not explore sparsity in intermediate layers, where it could interact differently with hierarchical feature learning.
- **Sweep limited to 3 points.** The levels tested (50%, 75%, 87.5%) leave room for interpolation. The regime between 87.5% and 100% (all information in a single dimension) is not explored and may reveal qualitatively different behavior.
- **Backprop end-to-end still used.** The k-WTA operation is applied in the forward pass but training uses standard SGD. Variants combining k-WTA with local plasticity rules are out of scope.
- **Comparison with 2017-era baselines.** ProtoNet remains a reference but more recent architectures (ResNet encoders, transformers) achieve higher accuracies on Omniglot. We do not position this work as state-of-the-art in raw accuracy—focus is on characterization of the sparsity-accuracy relationship in a widely studied backbone.

5.4 Future work

Some direct extensions: (i) finer sweep between $k = 8$ and $k = 16$ ($k \in \{4, 6, 10, 12\}$) to precisely characterize the threshold where degradation accelerates; (ii) multiple seeds (5–10) to obtain full training variability; (iii) k-WTA in intermediate layers, observing interaction with hierarchical feature learning; (iv) more difficult datasets (Mini-ImageNet, tieredImageNet) to test sparsity-accuracy generalization beyond Omniglot; (v) combination with meta-learned local plasticity rules.

5.5 Related observation: ProtoNet under continual learning

In parallel exploration in the same project, we tested ProtoNet in a sequential continual learning setting without replay (sequential training on 50 class groups, no buffer of old examples). We observed that the naive ProtoNet baseline—without any defense against catastrophic forgetting—reaches ∼80% average accuracy across 50 tasks, with moderate forgetting (BWT ≈ -9 p.p.). This robustness appears intrinsic to the prototype-based mechanism: since prototypes are computed FRESH at each support-set episode, there is no trained classifier head to "forget." Bio-inspired mechanisms we tested (meta-learned local plasticity, k-WTA in a combined architecture) did not exceed this baseline under our experimental conditions. Detailed documentation in supplementary material available upon request.

6 Conclusion

This paper presented a controlled empirical study of sparsity tolerance in Prototypical Networks applied to few-shot learning on Omniglot. Applying k-WTA with three sparsity levels (50%, 75%, 87.5%) to the final embedding of a standard CNN-4 and training end-to-end with gradient flow through the selected top-k channels, we observed that **75% of activations can be zeroed with a**

loss of only 1.45 percentage points in 5-way 1-shot accuracy (93.10% vs. 94.55% baseline). Validation with an untrained encoder confirms the gain comes from optimization under the constraint (37.60% without training vs. 93.10% trained), not the k-WTA structure alone.

The result contributes empirical evidence to the discussion of compatibility between biological sparse coding principles and deep representation learning. Although we do not argue for the biological fidelity of the tested mechanism, the result offers a concrete comparison point: prototype-based metric learning is not destroyed by explicit sparsity constraints, and the encoder learns to use sparse dimensions discriminatively when trained to do so.

Future work includes finer sweeps around the threshold observed between 75% and 87.5% sparsity, multi-seed validation, application in intermediate layers beyond the final embedding, and testing on visually more complex datasets.

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